7.1 Basic mapmaking

Before you explore the boundaries of mapping possibilities, you need to make a simple map. In D3, the simplest map you can make is a vector map using SVG `<path>` and `<circle>` elements to represent countries and cities. We can bring back cities.csv, which we used in chapter 2, and finally take advantage of its coordinates, but we need to look a bit further to find the data necessary to represent those countries. After we have that data, we can render it as areas, lines, or points on a map. Then we can add interactivity, such as highlighting a region when you move your mouse over it, or computing and showing its center.

Before we get started, though, let's take a look at the CSS for this chapter.

Listing 7.1 ch7.css

```
path.countries {
     stroke-width: 1;
     stroke: black;
     opacity: .5;
     fill: red;
}

circle.cities {
     stroke-width: 1;
     stroke: black;
     fill: white;
}

circle.centroid {
     fill: red;
     pointer-events: none;
}
rect.bbox {
     fill: none;
     stroke-dasharray: 5 5;
     stroke: black;
     stroke-width: 2;
     pointer-events: none;
}

path.graticule {
     fill: none;
     stroke-width: 1;
     stroke: black;
}

path.graticule.outline {
     stroke: black;
}
```

7.1.1 Finding data

Making a map requires data, and you have an enormous amount of data available. Geographic data can come in several forms. If you're familiar with GIS, then you'll be familiar with one of the most common forms for complex geodata, the *shapefile*, which